Sci Eng Ethics (2008) 14:1–2
DOI 10.1007/s11948-008-9054-9

EDITORIAL

Climate—An Item for the Ethics Agenda

Raymond E. Spier

Published online: 14 February 2008

Through this editorial, I would like to invite users of this journal to respond to the welter of published, spoken and dramatised material that is presently circulating concerning the climate of our planet.

I have to admit that I am uncomfortable with the consensus view of the Intergovernmental Panel on Climate Change (IPCC) that rising levels of carbon dioxide (CO_2) in the atmosphere have, through a greenhouse-type effect, caused the climate to increase in temperature by about 1.5°F since 1883 [1]. This overall change includes a period between 1943 and 1975 when the temperature fell by 0.4°F. While this temperature fall was ongoing, there was an increase in atmospheric CO_2 of 10 parts per million by volume (ppm) [2]. My second reason for unease is based on a viewing of the data of the Vostok ice-cores which provide temperature and CO_2 data for the last 450,000 years. Here it is clear that changes in temperature precede changes in CO_2 [3]. These data may be interpreted as having the temperature changes drive the changes in CO_2. Also for a rise in CO_2 of 1 ppm there is observed a change in temperature of 0.2°F which means that the global temperature should have risen 2°F during the 1943–1975 period instead of dropping 0.4°F. And the global temperature should be some 8°F hotter than it is today. Thirdly, the main greenhouse gas is water. Clouds constitute the primary control system on what temperatures are experienced on the surface of this planet. So the control of cloud formation is a probable determinant of global temperature. Some [4] hold that cosmic rays, whose quantity and nature are controlled by what the Sun emits and what the Earth's magnetic field directs, are implicated in the formation of clouds. While clouds reflect the Sun's light and keep the planet/things/it cool during the day, they also prevent heat from leaving the Earth and therefore keep it warm during the night.

R. E. Spier (✉)
University of Surrey, Guildford, Surrey, UK
e-mail: r.spier@surrey.ac.uk

Nevertheless, let me concede that the global temperature is rising. That this increase is in some way related to the amount of CO_2 in the atmosphere and that humans, by burning fossil fuels are responsible for this additional CO_2. What should be done? In his latest book, *Cool It*, Bjorn Lomborg makes the point that the increase in global temperature has beneficial effects as well as detrimental ones [5]. He also details the arguments that show beyond reasonable doubt that spending money on attempts to limit or lessen the amount of CO_2 in the atmosphere are not cost effective in achieving human benefits when the money could be spent otherwise. He cites spending on decreasing malaria, decreasing malnutrition, erecting better flood barriers, decreasing HIV/AIDS and providing clean drinking water and sanitation as areas where money can be spent with greater benefits to the humans of this planet. He also makes a strong plea for more, much more money to be spent on research on ways to achieve the satisfaction of an energy hungry humanity without burning irreplaceable fossil fuels. Lomborg identifies/cites wind, wave and nuclear power as able to provide some of these needs, although he strangely neglects to emphasise the transformation of solar energy to electricity using photovoltaic transformers. He also does not examine the effect of population growth on the use of Earth's limited non-renewable resources.

So, let me conclude somewhat provocatively by suggesting that the global community should let the CO_2 rise as it may and spend the money that the populace seems to want to devote to its stabilisation or reduction on research. At this level of expenditure (some $50 billion/annum), this would assure new ways of finding the energy to enhance standards of living without devouring fossil fuels; it might also provide us with political freedoms that we would appreciate greatly.

References

1. http://data.giss.nasa.gov/gistemp/graphs/ (accessed Jan 2007).
2. http://www.windows.ucar.edu/earth/Atmosphere/images/co2_concentration_1750_2000_big.gif (accessed Jan 2007).
3. Petit, J. R., Jouzel, J., Raynaud, D., Barkov, N. I., Barnola, J.-M., Basile, I., Bender, M., Chappellaz, J., Davis, M., Delaygue, G., Delmotte, M., Kotlyakov, V., Legrand, M., Lipenkov, V. Y., Lorius, C., PÉpin, L., Ritz, C., Saltzman, E., & Stievenard, M. (1999). Climate and atmospheric history of the past 420,000 years from the Vostok ice core, Antarctica. *Nature*, *399*, 429–436.
4. Courtillot, V., Gallet, Y., Le Mouël, J. L., Fluteau, F., & Genevey, A. (2007). Are there connections between the Earth's magnetic field and climate? *Earth and Planetary Science Letters*, 253(3–4), 328–339.
5. Lomborg, B. (2007). Cool it. New York: Knopf.